

Dr. Guangping Zeng

Postdoctoral Researcher

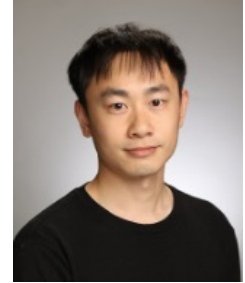
South China Sea Institute of Oceanology, Chinese Academy of Sciences

Deep Sea Drilling Center, Guangzhou Marine Geological Survey

Email: zengguangping22@mails.ucas.ac.cn

Research Interests

My research focuses on volatile (CO_2 and H_2O) cycling in subduction zones and its role in magma generation and metallogenesis. I use isotope geochemistry and thermodynamic modeling to constrain volatile cycling processes in the Makran system (an atypical low-angle subduction zone in the northern Arabian Sea with the thickest sediment inputs globally), and extend this work using big-data and machine learning approaches at the global scale.



Academic Appointments & Training

Postdoctoral Researcher (Advisor: Prof. Zhen Sun)	2025-present
South China Sea Institute of Oceanology, Chinese Academy of Sciences (affiliation)	
Deep Sea Drilling Center, Guangzhou Marine Geological Survey (host institution)	
Visiting Scholar (Advisor: Prof. Susanne M. Straub)	2024-2025
Lamont-Doherty Earth Observatory, Columbia University	
Ph.D., Marine Geology (Advisor: Prof. Yunying Zhang)	2022-2025
South China Sea Institute of Oceanology, Chinese Academy of Sciences	
M.S., Sedimentology and Paleogeography (Advisor: Dr. Jun Wang)	2019-2022
Institute of Sedimentary Geology, Chengdu University of Technology	

Research Experience

Makran Subduction Zone: Carbon Cycling	2024-present
Integrated petrology, isotope geochemistry, and thermodynamic modeling to constrain sedimentary carbon recycling mechanisms and fluxes in subduction zones.	
Cenozoic Volcanism in Pakistan: Petrogenesis and Tectonic Evolution	2022-2025
Applied geochronology, petrology, and geochemistry to investigate magma genesis and its tectonic implications.	
South Poyang Basin: Paleogeography and Organic Matter Enrichment	2019-2022
Combined sequence stratigraphy and geochemical analyses to reconstruct paleogeographic evolution and constrain controls on organic matter enrichment.	

Lab Skills

Geochemistry & Microanalysis: SIMS, MC-ICP-MS, LA-ICP-MS, SEM, XRF

Data Analysis: Python, MATLAB, ArcGIS

Honors and Awards

AGU Student Travel Grant \$1,000 (2024)
UCAS Joint Ph.D. Training Fellowship \$24,000 (2024-2025)
National Encouragement Scholarship (2020-2022)
Outstanding Graduation Thesis Award (2019)
Excellent Camper, Nanjing Institute of Paleontology, CAS (2018)
University People's Scholarship Award (2017)
University Student Summer Social Practice Group Award (2016)

Publications

Peer-reviewed Articles

G.P. Zeng, Y.Y. Zhang, Z. Sun, M. Mehmood, et al., 2026. Unraveling the India–Eurasia collision pattern: Insights from subduction- to collision-related magmatism in the Chagai Hills. *Journal of Earth Science*, *online first*.

S.A.H. Gardezi, X.W. Luan, Z. Sun, J. Lin, C.H. Zhang, **G.P. Zeng**, et al., 2026. Structural evolution of mud intrusive–extrusive systems in the Makran accretionary wedge: Processes and patterns. *Marine and Petroleum Geology*, *accepted*.

G.P. Zeng, J. Wang, C.G. Zhang, Q. Zhao, S. Wu, 2022. Paleogeographic environment characteristics of the Guanshan Member of the Leping Formation in the Erjiacun Depression, southern Poyang Basin. *Fault-Block Oil & Gas Field*, 29(2), 199–206 (*in Chinese*).

Conference Abstracts

G.P. Zeng, Y.Y. Zhang, S.M. Straub, et al., 2024. Sediment recycling at the Makran arc: Insights from whole-rock elemental and Sr–Nd–Pb isotope systematics. Abstract, AGU Fall Meeting 2024.

Manuscripts under Review

G.P. Zeng, Y.Y. Zhang, Z. Sun, et al. Efficient return of subducted organic carbon in the Makran arc volcano. *Under review*.